# NCFE Level 1 Functional Skills Qualification in Mathematics (603/5055/6)

Mark scheme: P001253
Assessment window: Monday 14 October 2019 – Friday 18 October 2019
v1.1

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## Examiner Mark Scheme Guidance

### Information

This guidance is intended to support NCFE examiners in the valid, reliable and consistent application of the relevant mark scheme version, against learner evidence generated during their external assessment.

This mark scheme provides:

- the total marks available for each question
- the subject content reference for each mark
- example process/methods and evidence of the types of responses expected for each mark
- (once confirmed) the pass mark for the relevant assessment version.

This mark scheme **must** be used for paper-based and online marking of the assessment version indicated**.**

### Instructions and guidance on application

- All learners must receive the same treatment and should be marked fairly. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Learners must be rewarded for what they have shown they can do rather than penalised for things they have not done.
- Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Half marks must not be awarded.
- Examiners should be prepared to award zero marks if the learner's response is not worthy of credit according to the mark scheme.
- The mark scheme is a working document and may be added to at the standardisation to reflect valid alternative answers given by a learner.
- When in doubt regarding the application of the mark scheme to a learner's response, the Chief Examiner must be consulted.

This mark scheme provides the following information:

- section and activity information
- question number
- total marks available per question (top row, shaded) followed by
- attribution of individual marks per question
- problem solving (PS) and underpinning skill (UPS) attribution
- process/method or answers, as well as additional or alterative evidence; indicative of the subject content (SC) attribution
- any additional guidance, as required.

To support the valid, reliable and consistent marking of learner evidence, the following abbreviations are applied throughout the mark scheme:

| Annotation | Explanation and use |
|---|---|
| **FT** | Follow through marks are applied when there are earlier mistakes in the method. |
| **OE** | Or equivalent marks are available for the justification of the answer being presented in a different form to the mark scheme i.e. 0.5 or ½. |

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| | |
|---|---|
| **CAO** | Correct answer only. |
| **Their** | 'Their' refers to the learners' own values. |
| **Seen** | Seen refers to the requirement to see the stated value in the learner's response or working out. |
| **Imp** | Implied refers to the learner's response implying correct working out used but not seen. |
| **Brackets** | Indicates units are not required on final answers or for answers seen within working. |
| **BOD** | Benefit of doubt where learner handwriting may be difficult to interpret but previous working may indicate correct final answer. |
| **Shaded** | Indicates requirements for full marks to be awarded. |

**Version Control**

Mark schemes are subject to version control. Examiners **must** ensure they have access to the latest version following each standardisation event.

Over time mark schemes will incorporate additional evidence captured and confirmed during standardisation events. Any additional evidence criteria will be captured in colour-coded text applicable to the dated standardisation event.

**Recording of marks**

*Paper-based:* Individual marks should be annotated in the 'Examiner' column in the learner script, added up and recorded at the end of each activity. The overall marks awarded for each learner should be clearly and legibly recorded in the grid on the front of the learner script.

*Online:* Onscreen marking tools (i.e. ticks, crosses) marks should be applied to indicate application throughout the learner script, in addition to marks being recorded numerically within the corresponding 'Learning Outcomes' box, indicated by the relevant Subject Content reference.

| Annotation | Explanation and use |
|---|---|
| **Tick** | Used to indicate correct values/method or final answer. |
| **Red highlight** | Used to indicate where the learner has made an error in either the value used or an incorrect calculation. |
| **Red line box** | Used to indicate where the learner may have made an error that has resulted in benefit of doubt being applied i.e. transposition of figures but previous working clearly shows otherwise. |

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**Note:** Pass marks for Functional Skills external assessments are set in an awarding meeting, in which a combination of statistical analysis and professional judgement is used to determine the minimum required standard to achieve a pass in the assessment.

While different versions of the same assessment are designed to be of the same level of difficulty, variations in content can lead to the minimum required standard being represented by different marks across versions.

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| Paper number: | Paper-based 1 (Oct 19) | Version: | 1.1 | Pass mark: | 37 |
|---|---|---|---|---|---|

**(Section A) Activity 1: Climbing mountains (Non-calculator Test)**

| Q | Marks | UPS / PS | Process and Answer | Additional or Alternative Evidence (with guidance) | SC |
|---|---|---|---|---|---|
| **1 (a)** | **2** | **PS** | 2015 | Award 2 marks if correct answer given | |
| | 1 | | 5 | 5 seen or indicated, i.e. 4 + 4 + 4 + 4 + 4 | N4 |
| | 1 | | 2015 | CAO | N2 |
| **1 (b)** | **1** | **UPS** | 7 hours 40 mins | CAO | M20e |
| **1 (c)** | **2** | **PS** | (£)26.40 | Award 2 marks if correct answer given | |
| | 1 | | $\frac{15}{100}\times 176$<br>or $\frac{176}{10}+\frac{1}{2}\left(\frac{176}{10}\right)$ or 17.6 + 8.8 | OE Correct method for finding 15% of 176 | M19 |
| | 1 | | (£)26.40 | CAO<br>Must be to 2 dp | M19 |
| **1 (d)** | **5** | **PS** | 6 (hours) | Award 5 marks if correct answer given | |
| | 1 | | 12 × 17 or 204 | | N5 |
| | 1 | | (1345 ÷ 100) × 10 or 134.5 | OE | N3b |
| | 1 | | 338.5 | | N5 |
| | 1 | | **their** 338.5 ÷ 60 or 5.6(4…) or 5 hours and 38.5 minutes | Not 5.3 hours<br>"Their" is from a correct addition of first two values | N11b |
| | 1 | | 6 (hours) | CAO | M20e |
| **1 (e)** | **1** | **UPS** | Option 3 - The angle is more than 12º but less than 90º | CAO | M24c |
| **1 (f)** | **2** | **UPS** | 9 9 (degrees) | Award 2 marks if correct answer given | |
| | 1 | | 3 – (–6) or 3 + 6 | Allow attempt to count along number line with 3 and – 6 marked<br>May be evidenced by –9 | N2 |
| | 1 | | 9 (degrees) | CAO (do not accept –9) | N2 |
| **1 (g)** | **2** | **PS** | See below | | |

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|---|---|---|---|---|---|
| | 1 | | $\frac{2}{10}$ or 30% seen | Accept 0.2 **and** 0.3 or $\frac{20}{100}$ **and** $\frac{30}{100}$ | N16 |
| | 1 | | Rain | Must be supported by comparison of numbers in same form | H30b |

| **(Section B) Activity 2: Community nurse (Calculator Test)** | | | | | |
|---|---|---|---|---|---|
| **Q** | **Marks** | **UPS / PS** | **Process and Answer** | **Additional or Alternative Evidence (with guidance)** | **SC** |
| **2 (a)** | **4** | **PS** | See below | | |
| | 1 | | 3 or more non-overlapping class intervals (e.g. 0-9,10-19, 20-29, 30-39 – see table below) | Allow intervals of different sizes<br>Allow if additional (unnecessary) classes added | H28a |
| | 1 | | Frequencies total 24 | Award mark if individuals are incorrect but total 24 | H28a |
| | 1 | | Frequencies correct | | H28a |
| | 1 | | Suitable headings for table | For example, (Number of) minutes and (number of) people. Do not award if "Time" given for first column | H28a |

**Additional guidance**

Example

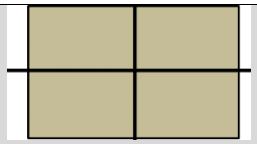
| Minutes | | | Frequency |
|---|---|---|---|
| 0 – 9 | or | 1 – 10 | 8 |
| 10 – 19 | or | 11 – 20 | 9 |
| 20 – 29 | or | 21 – 30 | 2 |
| 30 – 39 | or | 31 – 40 | 5 |

| Min | Freq |
|---|---|
| 2 | 1 |
| 3 | 1 |
| 4 | 1 |
| 5 | 1 |
| 6 | 1 |
| 7 | 1 |
| 8 | 2 |
| 11 | 2 |
| 12 | 3 |
| 13 | 1 |
| 14 | 1 |
| 15 | 1 |
| 18 | 1 |
| 25 | 1 |
| 28 | 1 |
| 31 | 1 |
| 32 | 2 |
| 35 | 1 |
| 37 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| **2 (b)** | **3** | **PS** | See below | | |
| | **Alternative method 1** | | | | |
| | 1 | | $\frac{40}{100} \times 15$ or $1.5 \times 4$ | OE correct method for finding 40% of 15 | N14 |
| | 1 | | 6 | Dep on calculation seen. | N14 |

| | | | | | |
|---|---|---|---|---|---|
| | 1 | | No and 8, supported by correct working | | N13 |
| | **Alternative method 2 (not in spec for L1 but may be seen)** | | | | |
| | 1 | | 8 | 8 from graph | N14 |
| | 1 | | $\frac{8}{15} \times 100 = 53$ | OE correct method for showing that 8 is 53% of 15 | N14 |
| | 1 | | No, supported by correct working | | N13 |
| **2 (c)** | **3** | **PS** | 13 and (£)3.64 in table | Award 3 marks if correct answer given | |
| | 1 | | 13 (km) in table | | M21 |
| | 1 | | **their** 13 × 28 or 364<br>or **their** 13 × 0.28 or 3.64 | "**Their**" is from the value calculated in 1st mark | N17 |
| | 1 | | 13 and (£)3.64 in table | CAO | M20d |
| **2 (d)** | **1** | **UPS** | 180 (°) | CAO | M26a |
| **2 (e)** | **2** | **PS** | See below | | |
| | **Alternative Method 1** | | | | |
| | 1 | | 800 ÷ 52 or 15.38(…) or 15.4<br>or 52 × 17 or 884 | | N17 |
| | 1 | | 15 or 15.38(…) or 15.4 or 884<br>and no | CAO<br>Award 2 marks if correct answer given | N17 |
| | **Alternative Method 2** | | | | |
| | 1 | | 800 ÷ 17 or 47.05(…) or 47.06 | | N17 |
| | 1 | | 47.05(…) or 47.06 and no | CAO<br>Award 2 marks if correct answer given | N17 |
| **2 (f)** | **2** | **UPS** | See below | | |
| | 1 | | 40 × 130 (=5200) | | N15 |
| | 1 | | 52 or 5270 or (£) 52.70 seen | **No** FT on **their** 5200 | M20d |

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| Activity 3: French flag (Calculator Test) | | | | | |
|---|---|---|---|---|---|
| **Q** | **Marks** | **UPS / PS** | **Process and Answer** | **Additional or Alternative Evidence (with guidance)** | **SC** |
| **3 (a)** | **2** | **UPS** | 7.5 (m) | Award 2 marks if correct answer given | |
| | 1 | | 225 + 225 + 150 + 150 or 750<br>or 2.25 + 2.25 + 1.5 + 1.5 or 7.5 | | M22b |
| | 1 | | 7.5 (m) | CAO | M20a |
| **3 (b)** | **1** | **UPS** |  | Mark intention. Both lines of symmetry required. | M24b |
| **3 (c)** | **4** | **PS** | 937 or 938 (plants) | Award 4 marks if correct answer given | |
| | 1 | | 225 × 150 or 75 × 150<br>or 33750 or 11250 | Method for finding area of flag or stripe | M22a |
| | 1 | | 11250 | CAO | M22a |
| | 1 | | **their** 11250 ÷ 12 or 937.5 | FT only from **their** area calculation in 1st mark | N17 |
| | 1 | | 937 or 938 (plants) | FT **their** plants correctly rounded to a whole number from 3rd mark | N12 |
| **3 (d)** | **2** | **UPS** | (£)35.96 | Award 2 marks if correct answer given | |
| | 1 | | (42.98 + 31.56 + 33.98 + 42.99 + 35.75 + 28.50) ÷ 6 or 215.76 ÷ 6 | | H29a |
| | 1 | | (£)35.96 | | H29a |
| **3 (e)** | **2** | **PS** | 11.6p or £0.116 and yes | Award 2 marks if correct answer given | |
| | 1 | | **their** 35.96 ÷ 310 or 3596 ÷ 310<br>or **their** 215.76 ÷ (310 × 6)<br>or 21576 ÷ (310 × 6)<br>or (£) 0.116 or 11.6 (p) | FT from "**Their**" final value for Q3d | H29a |
| | 1 | | 11.6p and yes<br>or £0.116 and £0.01 and yes | Accept rounding to 12p | M20d |
| **3 (f)** | **2** | **PS** | 6 (litres) | Award 3 marks if correct answer given | |
| | 1 | | 30 × 50 × 4 or 6000 ($cm^3$) | | M23 |
| | 1 | | 6 (litres) | CAO | M20c |

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|---|---|---|---|---|---|
| **3 (g)** | **2** | **PS** | 4.2 × 4 = 16.8 and yes **OR** 21:84 = 1:4 and yes | Award 2 marks if correct answer given | |
| | 1 | | 4.2 and 16.8<br>OR $\frac{21}{5}$ and $\frac{84}{5}$ OR 21:84 | | N8b |
| | 1 | | 4.2 × 4 = 16.8 OR 21:84 = 1:4 and yes | CAO, accept equivalent correct comparisons | N17 |

| **Activity 4: World hunger (Calculator)** | | | | | |
|---|---|---|---|---|---|
| **Q** | **Marks** | **UPS / PS** | **Process and Answer** | **Additional or Alternative Evidence (with guidance)** | **SC** |
| **4 (a)** | **2** | **PS** | See below | | |
| | **Alternative method 1** | | | | |
| | 1 | | 7726.5 ÷ 9 or 7726 500 000 ÷ 9 | | N9 |
| | 1 | | 859 (million) | CAO | N12 |
| | **Alternative method 2** | | | | |
| | 1 | | 7726.5 x 0.11 or 849.91(…) | | N9 |
| | 1 | | 850 (million) | CAO | N12 |
| **4 (b)** | **1** | **UPS** | 39 | CAO | H29b |
| **4 (c)** | **2** | **UPS** | 0.3 | Award 2 marks if correct answer given | |
| | 1 | | $\frac{6}{20}$ or $\frac{3}{10}$ | OE | H31 |
| | 1 | | 0.3 | CAO | N16 |
| **4 (d)** | **3** | **PS** | 25 | Award 3 marks if correct answer given | |
| | 1 | | 1.60 × 1.60 or 2.56 | | N5 |
| | 1 | | 64 ÷ **their** 2.56 | FT **their** 2.56 from first mark | N5 |
| | 1 | | 25 | CAO | N5 |
| **4 (e)** | **2** | **PS** | See below | | |

| | | | | | |
|---|---|---|---|---|---|
| | 1 | | 0.2 × 300 or (300 ÷ 10) × 2 or 60<br>or 0.15 × 440 or $\frac{440}{10}+\frac{440}{10}\times\left(\frac{1}{2}\right)$ or 66 | OE Any full correct method for finding 20% of 300 or 15% of 450 | N14 |
| | 1 | | 60 and 66 and Village B | With correct supporting working | N13 |
| **4 (f)** | **1** | **UPS** | 1 000 000 | CAO Ignore commas.<br>Do not allow full stops/decimal points. | N1a |
| **4 (g)** | **2** | **PS** | $\frac{3}{20}$ | Award 2 marks if correct answer given | |
| | 1 | | $\frac{15}{100}$ | | N16 |
| | 1 | | $\frac{3}{20}$ | CAO | N16 |
| **4 (h)** | **2** | **PS** | See below | | |
| | 1 | | All sectors correct size but labels (or key for onscreen version) missing<br>or 2 sectors correct and correctly labelled. | Supermarkets 3 segments OR 54 (degrees)<br>Restaurants 7 segments OR 126 (degrees)<br>Households 9 segments OR 162 (degrees)<br>Others 1 segment OR 18 (degrees) | H27b |
| | 1 | | All sectors correct and labelled (or key for onscreen version) | No tolerance | H27b |

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